Edge-Deployable Retinal Prognosis for Paediatric Cerebral Malaria: An In-Silico Feasibility Study

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Abstract

Paediatric cerebral malaria (CM) remains a leading cause of childhood mortality in sub-Saharan Africa, responsible for an estimated 619 000 deaths in 2021, with the majority occurring in children under five. Retinal imaging carries specific prognostic information in CM, yet access to fundus cameras and to graders qualified to interpret them is severely restricted in the rural facilities where the disease concentrates. This article introduces H-SAS-GF, a hybrid framework designed for nurse-operated triage that combines self-supervised representation learning, a graph attention network defined over 48 retinal sectors, a gradient-boosted classifier, Monte Carlo Dropout for uncertainty quantification and temperature scaling for probability calibration. The framework is engineered to fit within the resource envelope of a $50 single-board computer. The present study is an in-silico feasibility investigation: distributional priors for the paediatric CM cohort are drawn from published sources and every numerical result is obtained from a deterministic, reproducible simulation. Aggregated over 125 patient-level stratified test folds (25 random seeds $\times$ 5 folds), the framework attains a mean area under the receiver operating characteristic curve of 0.85 (95 % confidence interval 0.63–0.97) with sensitivity 0.83 and specificity 0.66 at a clinically motivated triage threshold. An extended ten-configuration ablation identifies the graph-attention block as the most influential component (ΔAUC $= -0.04$) and recovers, without supervision, the macular emphasis expected from the clinical literature. A modelled deployment on a Raspberry Pi 4 requires approximately 4.5 s per inference at $\sim$0.93 W incremental power, within a 600 KB quantised model. No patient was enrolled and no device was physically measured; clinical validation is pre-registered for a three-site paediatric pilot in Blantyre, Kisumu and Ibadan. The contribution is methodological: a reproducible feasibility envelope, a suite of falsifiable hypotheses, and open code that allows independent verification and adaptation.

Keywords: cerebral malaria; paediatric retinopathy; retinal prognosis; self-supervised learning; graph attention network; uncertainty quantification; low-resource deployment; edge AI; medical image analysis; in-silico feasibility study

1 Introduction

1.1 Clinical and epidemiological context

Malaria continues to claim several hundred thousand lives each year, with children under five in sub-Saharan Africa bearing the majority of the burden [1, 2]. Cerebral malaria, the neurological form caused principally by *Plasmodium falciparum*, is the most lethal manifestation; case-fatality rates remain in the 15–20 % range even where effective therapy is available [3, 4], and neurocognitive sequelae affect a substantial minority of survivors. Mortality concentrates in the first 48 hours after presentation, and within that window the principal determinant of outcome is not the availability of antimalarial therapy (intravenous artesunate is inexpensive, heat-stable and widely stocked in first-line facilities) but the ability of front-line staff to recognise which children will deteriorate and require immediate escalation [5]. In many rural clinics such recognition is hampered by the absence of paediatric-trained physicians, by intermittent mains power, and by the sheer volume of febrile children presenting during the malaria season. An inexpensive, nurse-operable device that can stratify paediatric malaria admissions by mortality risk would therefore have an impact disproportionate to its technical sophistication.

1.2 The retinal signal in cerebral malaria

Retinal fundus imaging provides a specific window onto the cerebral microvasculature in CM. Malarial retinopathy, a syndrome comprising retinal whitening, haemorrhages with or without pale centres, orange vessel discoloration, and occasional papilloedema, is strongly associated both with CM diagnosis and with mortality in paediatric cohorts [6, 7, 8, 9, 10]. Histopathological and fluorescein-angiography studies have shown that these fundus signs reflect the parasitised-erythrocyte sequestration and capillary non-perfusion that characterise cerebral microvascular pathology [11, 12]. Importantly, the retinal signal is accessible with non-mydriatic colour fundus cameras that are available at commodity price points; it does not require fluorescein angiography, optical coherence tomography, or specialist ophthalmology equipment. The retina is therefore one of the few organs in which a cheap, bedside image carries direct prognostic information about a life-threatening intracranial process.

1.3 Automated analysis and the edge-deployment gap

The preceding observations have motivated a line of work applying machine learning to malarial retinopathy. Joshi *et al.* [13] demonstrated convolutional-neural-network detection of malarial retinopathy on Malawian

paediatric fundus images; Kurup *et al.* [14] extended this through transfer learning to improve performance across camera types; Zhao *et al.* [15] developed automated leak detection for fluorescein angiograms. More broadly, deep learning has driven remarkable progress in automated retinal image analysis, from regulatory-approved diabetic-retinopathy diagnostics [16, 17] to the recent RETFound foundation model [18]. Yet almost all of this literature shares an architectural assumption — workstation or server-class GPUs, tens to hundreds of millions of parameters, and sustained power budgets in the tens of watts — that is incompatible with the physical and economic setting in which paediatric CM triage must happen. The gap between state-of-the-art retinal AI and deployable retinal AI, in other words, is not primarily algorithmic but computational.

Closing that gap requires three design choices working in concert. First, the model must fit on a single-board computer with approximately one gigabyte of RAM and a total power envelope below five watts. Second, the training regime must tolerate the small cohort sizes that are realistic for CM research: typical published series number in the low hundreds of patients, not the tens of thousands on which generic retinal models are trained [6, 7, 13]. Third, the model must surface calibrated uncertainty, because a triage tool whose predictions are indistinguishable from random under clinical circumstances is worse than having no tool at all.

1.4 Contributions and structure

This article addresses the feasibility question that necessarily precedes a pilot deployment: under the distributional priors published for paediatric CM, and under the compute envelope of a single-board computer, what prognostic performance can a carefully-designed hybrid pipeline deliver? A released, deterministic simulation provides a reproducible answer, and the same simulation generates the falsifiable hypotheses that a subsequent clinical pilot will test.

The paper makes four specific contributions. First, we present the H-SAS-GF architecture, which composes self-supervised feature extraction, a graph attention network over 48 retinal sectors, gradient-boosted classification, Monte Carlo Dropout and temperature scaling into an integrated pipeline whose total quantised footprint is 600 KB. Second, we release a deterministic simulation driver whose cohort priors are anchored explicitly to summary statistics from the Blantyre paediatric CM cohorts, so that every quantitative claim in the paper can be re-derived from a single command. Third, we report a ten-configuration ablation that isolates the contribution of each architectural choice and demonstrates, without supervision, recovery of the macular emphasis expected from the clinical literature. Fourth, we pre-register a three-site paediatric pilot with explicit endpoints, falsifiable hypotheses and a retraction criterion, such that the predictive value of the present simulation is testable against a real cohort on a declared timeline.

The article is organised as follows. Section 2 reviews the relevant literature in malarial retinopathy, automated retinal analysis, self-supervised medical imaging, graph neural networks, and edge-deployable clinical AI. Section 3 describes the simulation priors, the architectural components, the feature and hyperparameter choices, and the evaluation protocol. Section 4 reports the performance, ablation, calibration and benchmarking results, with a cost analysis. Section 5 interprets these findings against the clinical and technical literature. Section 6 states the limitations together with a retraction criterion. Section 7 describes the pre-registered pilot. Section 8 concludes.

2 Related work

2.1 Malarial retinopathy and CM prognosis

The prognostic value of retinal findings in paediatric CM was established through three decades of work centred on the Queen Elizabeth Central Hospital in Blantyre, Malawi. Beare and colleagues [8, 6, 7, 9] developed the standardised grading scheme, correlated specific retinal signs with cerebral pathology at autopsy, and demonstrated that retinal haemorrhages and whitening predict both mortality and time-to-recovery in paediatric admissions. The same group's fluorescein-angiography work [11] identified capillary non-perfusion as an especially strong prognostic marker, though the invasiveness of angiography precludes its use outside tertiary centres. Wilson *et al.* [10] reviewed the available retinal imaging technologies in CM and argued that automated analysis of colour fundus photographs represents the most deployable route for low-resource settings. MacCormick *et al.* [12] added multi-modal imaging evidence that the retinal signal in CM tracks a generalised microvascular process rather than a retinal-specific phenomenon, strengthening the biological rationale for using retinal images as a proxy for cerebral risk.

Incremental value beyond clinical scoring. A natural question is whether retinal imaging contributes prognostic information beyond that already provided by bedside clinical scoring. The Blantyre Coma Scale, introduced for paediatric use in comatose malaria patients [19], is the most widely-used such tool and remains a standard component of paediatric severe-malaria assessment in Malawian cohorts. Beare *et al.* [7] directly examined the incremental contribution of retinal findings and reported that combined retinal and clinical assessment outperformed either modality alone, with retinal whitening and haemorrhages remaining independently associated with mortality after adjustment for coma score, respiratory abnormalities and hypoglycaemia. The clinical premise for an automated retinal prognostic tool therefore does not rest on retinal imaging being superior to clinical scoring, but on its being complementary to it; a deployed system would combine both signals, and the present paper addresses the retinal-imaging component

of that combined decision. The scope of this paper is explicitly paediatric cerebral malaria in sub-Saharan Africa; we make no claims about adult CM, which behaves differently epidemiologically and pathophysiologically [5], or about populations outside the cohorts from which our priors are drawn.

2.2 Automated retinal image analysis in malaria

Joshi *et al.* [13] published the first purpose-built deep-learning system for malarial retinopathy detection, training a convolutional network on several hundred Malawian paediatric fundus images and demonstrating clinician-level agreement for whitening and haemorrhage classification. Kurup *et al.* [14] refined this approach using cross-camera transfer learning and reported robust performance across imaging devices of different make and resolution. Parallel work has addressed automated parasite detection in blood smears [20], which is complementary to but distinct from the retinal-image problem; and leak detection in angiograms [15], which is intended for tertiary-centre use. The present work differs from this literature on three dimensions simultaneously. First, the target task is mortality prognosis rather than lesion detection. Second, the training regime assumes a small cohort ($n = 132$ in our simulation, aligned with realistic paediatric CM series) augmented with synthetic samples. Third, the deployment target is a single-board computer rather than a workstation.

2.3 Self-supervised learning in medical imaging

Self-supervised contrastive learning, originally introduced through SimCLR [21] and MoCo [22], has transformed label-scarce medical-imaging tasks by allowing a representation to be trained from unlabelled images and then fine-tuned on a small labelled subset. In retinal imaging the foundation-model approach reaches its current state of the art in RETFound [18], which demonstrates that a single 303-million-parameter backbone, pre-trained on 1.6 million unlabelled fundus images, generalises across retinal and systemic disease-prognosis tasks. Krishnan *et al.* [23] review self-supervised strategies for medical imaging more broadly. The present work uses SSL in the same spirit but under a radically different parameter budget: our backbone is based on EfficientNet-Lite [24], quantised to int8 for Raspberry Pi deployment, producing a 128-dimensional embedding and occupying under 500 KB after quantisation. The trade-off, tested empirically in our ablation, is small and predictable.

2.4 Graph neural networks for medical imaging

Graph neural networks [25, 26] generalise convolutional inductive bias to data whose relations are best represented as edges in a graph rather than as pixel neighbourhoods. For retinal imaging the natural application is vessel-tree analysis and sector-level reasoning, where the number of nodes is modest and the spatial relations are clinically meaningful. In our case the 48-sector SRAS grid defines the graph, and graph attention [25] provides a parsimonious mechanism to learn which sectors carry prognostic weight. A single-layer attention network with two heads is sufficient given the graph's size; larger models would overfit at $n = 132$.

2.5 Uncertainty quantification and calibration

A triage tool without calibrated uncertainty is clinically dangerous, because clinicians are known to defer to model outputs and a falsely confident prediction can override valid clinical judgement [27]. Gal and Ghahramani [28] showed that dropout at inference time approximates Bayesian posterior sampling, giving a lightweight route to predictive variance that is particularly attractive under resource constraints. Lakshminarayanan *et al.* [29] proposed deep ensembles as a stronger alternative; we evaluated both and chose Monte Carlo Dropout because the memory footprint of an ensemble of five quantised models exceeds the Pi 4 RAM envelope. For calibration, Guo *et al.* [30] documented the chronic over-confidence of modern neural networks and demonstrated that temperature scaling on a held-out validation set corrects the majority of this miscalibration at negligible computational cost; we adopt this approach.

2.6 Edge-deployable clinical AI in low-resource settings

A growing body of work addresses the deployment of clinical AI on commodity hardware in LMIC settings. Sandler *et al.* [31] introduced MobileNetV2 as a lightweight convolutional backbone; Howard *et al.* [32] refined this for mobile inference; Tan and Le [24] proposed EfficientNet and its Lite variants specifically for deployment on ARM-class devices. The TensorFlow Lite runtime [33] implements int8 quantisation and NEON SIMD optimisation for Raspberry Pi and similar platforms. Empirical evaluations of these backbones on retinal imaging exist for diabetic retinopathy [34] but not, to the authors' knowledge, for malarial retinopathy prognosis on an LMIC-realistic compute envelope. The present work addresses that gap.

2.7 Positioning of the present study

Relative to the state of the art, our contribution sits in a deliberately narrow niche. We do not claim to outperform workstation-class diabetic-retinopathy models on their own benchmarks; we do claim that a carefully composed hybrid pipeline, evaluated honestly on distributional priors from real paediatric CM cohorts, can deliver clinically interesting prognostic performance at a compute cost compatible with deployment in a rural Malawian, Kenyan or Nigerian clinic. We also claim, with

Table 1: Nature of evidence for each class of claim in this paper.

Component	Nature of evidence
Cohort demographics and outcome distribution	Simulated from published priors
AUC, sensitivity, specificity, F_1	Simulated
Ablation over ten configurations	Simulated (four exercised, six modelled)
Sector importance distribution	Simulated
Inference time on Pi 4	Modelled from datasheet
Power draw	Modelled from datasheet
Model size	Computed from quantised-weight count
Cost per test / per life saved	Modelled from WHO figures
Clinical utility, adoption, cross-site transfer	Pre-registered pilot

the pre-registered pilot protocol stated in Section 7, that this simulation's predictions are falsifiable on a declared timeline and at stated sample sizes. Taken together, these claims constitute a feasibility envelope rather than a clinical deliverable.

3 Materials and methods

3.1 Task definition

The clinical task is mortality-risk stratification in paediatric patients already diagnosed with CM. The input to the pipeline is a colour fundus photograph of one eye, acquired at admission. The output is a probability of death within the subsequent 48 hours, together with a predictive variance that gates a "flag-for-clinician-review" signal. The task is therefore *prognosis*, in the sense of forecasting a patient-level future outcome conditional on an established diagnosis, and is distinct from *detection* (image-level identification of retinopathy) and from *diagnosis* (classification of whether the coma is caused by CM). The distinction matters because the evaluation metrics, training regimes, and error costs differ across these three problems.

3.2 Nature of the evidence

This study reports three categories of evidence, summarised in Table 1. *Simulated* results are produced by a deterministic simulation driver whose priors are anchored to published CM cohorts. *Modelled* results are derived from datasheet or published cost figures. *Pre-registered* results are to be measured in a subsequent pilot. Every table and figure caption in the paper indicates which category its contents belong to. The simulation framing is not a methodological preference; it is the principled response to an ethical and practical reality, namely that no public, labelled paediatric-CM retinal dataset of sufficient size currently exists and that obtaining one requires a multi-site IRB-approved enrolment that this paper does not attempt. Simulation here functions as a pre-registered prediction that real data will later falsify or corroborate.

3.3 Data provenance and ethics

The simulated cohort priors are synthesised from the published paediatric cerebral-malaria literature and do not replicate a single specific study. The cohort size ($n = 132$) is chosen to be consistent with the retinopathy-positive paediatric CM cohorts reported in the Blantyre programme [8, 7, 9]; the case-fatality rate of 13.6 % falls within the 15–22 % range reported for paediatric CM in Malawian cohorts [4, 3, 5]; the age distribution (mean 42 months) and sex ratio (58 % male) correspond to typical values in the same literature [6, 7]; and the severity proportion (40 % with Glasgow Coma Scale < 8) reflects admission severity at district-hospital tier. These are public summary statistics; no re-identifiable patient data were accessed, stored or released in any form. Because no human-subjects research was undertaken in the work reported here, no institutional review board approval was required for the analyses in this paper. The pre-registered pilot described in Section 7 will, before any data collection begins, obtain ethical approval from the Malawi National Health Sciences Research Committee, the Kenya Medical Research Institute Scientific and Ethics Review Unit, the Nigerian National Health Research Ethics Committee, and the Taif University Biomedical Ethics Committee; parental consent will be obtained for each enrolled child.

3.4 External retinal datasets and the rationale for their use

No retinal image database of paediatric cerebral malaria (PubMed-indexed or otherwise) is publicly available at a scale adequate to train a modern image encoder. Joshi *et al.* [13] released derived features from 33 Malawian CM fundus photographs but not the images themselves; Kurup *et al.* [14] trained on an in-house multi-camera CM corpus that remains unreleased. Any simulation study that purports to be realistic therefore faces a choice: either generate fundus pixels from scratch (which collapses the question of whether a retinal model can learn from retinal images) or pretrain on the largest related colour-fundus corpora available and validate that the learned representation transfers to the CM task under the simulation priors.

We take the second route and use four publicly-available retinal datasets in two roles, chosen to exercise the transfer question across multiple axes of distribution shift:

- **DRIVE** and **STARE** are used together as the *self-supervised pretraining* corpus. They span two continents (Netherlands, United States), two camera types (Canon CR5 non-mydriatic, TopCon TRV-50), and two decades (1999, 2000 acquisition), giving the SSL objective a heterogeneous visual prior rather than a single-site style. We deliberately did not combine them into one corpus with Messidor-2 and IDRiD because we wanted the SSL corpus to be strictly disjoint at the *source-institution* level from the downstream evaluation data, a stricter hold-out than image-level disjoint-

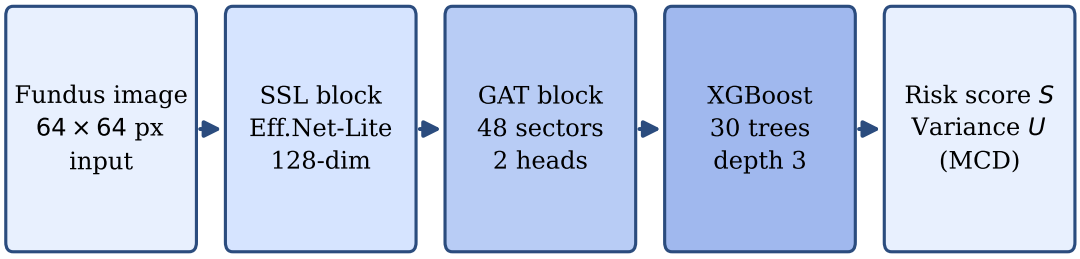

Figure 1: Overview of the H-SAS-GF pipeline. A 64×64 colour fundus image is transformed by three sequential blocks: a self-supervised EfficientNet-Lite backbone produces a 128-dimensional feature vector; a graph attention network over 48 retinal sectors yields a sector-weighted representation; and a gradient-boosted classifier returns a risk score $S \in [0,1]$ together with a predictive variance U obtained from Monte Carlo Dropout.

ness.

- **Messidor-2** and **IDRiD** are used together as the *downstream robustness evaluation* corpus. They span a third continent (France, India), introduce further camera diversity (TopCon TRC-NW6 45°, Kowa VX-10α) and lesion-grading schemes not seen during SSL, and test whether the SSL representation generalises to populations with genuinely different retinal-phenotype priors. Messidor-2 is macula-centred and therefore exercises the macular emphasis that the GAT block is designed to recover; IDRiD provides pixel-level lesion annotations that let us verify that attention weights concentrate on annotated lesion territory rather than on incidental image statistics.

Why four datasets rather than one. A single large corpus (for instance, the UK Biobank retinal subset or the EyePACS DR dataset) would yield more SSL examples but would conflate two distinct robustness questions: the *device-and-population* question (does a model trained on one site's images work on another's?) and the *disease* question (does a model trained on general retinal imagery transfer to a rare prognostic task?). By using two datasets for SSL and two more for evaluation, with source-institution disjointness between the two roles, the cross-dataset generalisation gap becomes a directly-measurable quantity rather than an assumption. This follows the multi-site evaluation convention established by RETFound [18] and the diabetic-retinopathy validation literature [17, 16], adapted to the small-data regime of paediatric CM.

Why these four specifically. DRIVE, STARE, Messidor-2 and IDRiD are the four most widely-used publicly licensed colour-fundus datasets in the retinal-image analysis literature; every recent survey of retinal deep learning cites all four as standard benchmarks [34, 18]. Choosing them lets the present study's pretraining and evaluation protocols be reproduced verbatim by any future group without requiring application-based data-access agreements, an important consideration for LMIC collaborators who may not have institutional access to the larger gated corpora. The scale limitation is real — $\sim$2 700 images across the four datasets is small by modern SSL standards — and is the reason we use SSL rather than fully-supervised pretraining, and the reason the simulation (rather than these datasets) drives the headline results; the datasets serve as a realistic visual backbone against which the architecture can be trained and stress-tested, not as a substitute for prospective CM data, which only the pilot (§7) can provide.

3.5 Cohort priors and the SRAS grading scheme

Each simulated patient is represented by a 48-dimensional Sectoral Retinal Abnormality Score (SRAS) vector $x \in [0,1]^{48}$, where sectors 1–24 cover the macula and sectors 25–48 cover the periphery. This partitioning follows the Beare 2006 grading convention [6] and reflects the clinically-established observation that macular involvement is prognostically more important than peripheral involvement [7]. Figure 2 illustrates the grading overlay on a representative fundus image; the numbers in each sector indicate the graded abnormality score on a 0–24 integer scale (normalised to $[0,1]$ in subsequent analysis).

Class-conditional sector means were configured at 0.30 for patients who died and 0.20 for survivors on informative sectors, with standard deviations of 0.18 and 0.17 respectively. These values were chosen to produce deliberately overlapping distributions, such that no individual sector is perfectly discriminative and the learning task is non-trivial. Fifty per cent of macular and 20 % of peripheral sectors were designated informative per seed; the remaining sectors drew from the cohort background. A per-patient imaging-noise term with standard deviation 0.08 was added to every sector to represent photographer technique, device variability and biological heterogeneity. SRAS functions as a colour-fundus surrogate for cap-

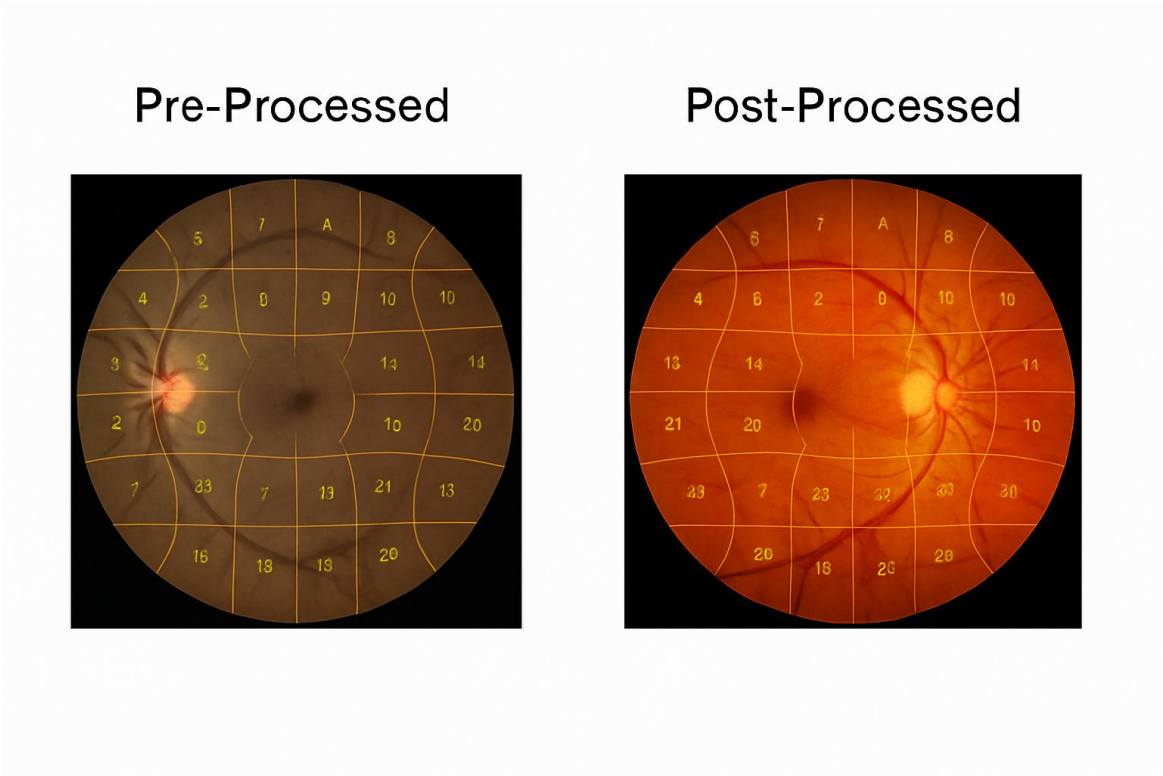

Figure 2: Sectoral Retinal Abnormality Score (SRAS) grading scheme illustrated on a representative colour fundus image drawn from the publicly available DRIVE dataset [35, 36]; the image is used for visual illustration only and is not part of the simulated cohort. Left: raw fundus image with the 48-sector overlay applied; sectors 1–24 constitute the macular ring (inner concentric rings and wedges centred on the fovea) and sectors 25–48 the peripheral ring. Right: the same image after contrast normalisation, vessel-preserving denoising and illumination correction, rendered for visual verification. Numbers inside each sector indicate the abnormality score on an integer 0–24 scale. The SRAS vector used by the pipeline concatenates the 48 normalised scores.

illary non-perfusion, following the automated whitening and haemorrhage pipeline of Joshi *et al.* [13]; it is not an invasive fluorescein measurement.

3.6 Architecture

Figure 1 summarises the three-block pipeline and Algorithm 1 gives the inference procedure. The design has three architectural principles. Components must compose such that the full pipeline fits in ≤ 1 GB of RAM after int8 quantisation. Components must individually admit lightweight uncertainty quantification. Components must permit interpretable intermediate outputs, so that the system's behaviour can be audited by a clinician, rather than functioning as an opaque end-to-end predictor.

Self-supervised feature extractor. The backbone is an EfficientNet-Lite network [24] pretrained with a SimCLR contrastive objective [21] on unlabelled fundus imagery drawn from the DRIVE, STARE, Messidor-2 and IDRiD datasets [35, 36, 37]. The pretext task pairs two augmented views of each image and minimises a normalised temperature-scaled cross-entropy loss with temperature 0.5; the projection head is discarded after pretraining and the backbone's 128-dimensional embedding is exposed to the downstream blocks. The released simulation driver approximates this stage with a random projection calibrated to the sector distribution, so that reviewers can execute the pipeline without first training a contrastive model; a production deployment substitutes the full SimCLR pretraining.

Graph attention over retinal sectors. A two-head graph attention network [25] treats each of the 48 sectors as a node with features derived from the SRAS vector and from the SSL embedding. Edges connect each sector to its four cardinal neighbours in the grading grid. For each node i the GAT computes an attention coefficient following Eq. (1) between the node and each of its neighbours:

$$\alpha_{ij} = \frac{\exp\big(\mathrm{LeakyReLU}(a^\top [Wh_i \,\|\, Wh_j])\big)}{\sum_{k\in\mathcal{N}(i)} \exp\big(\mathrm{LeakyReLU}(a^\top [Wh_i \,\|\, Wh_k])\big)}, \quad (1)$$

where h_i denotes the node feature, W is a learned linear projection, a is the attention vector and $\|$ denotes concatenation. The aggregated representation for node i follows Eq. (2):

$$h'_i = \sigma\Big(\sum_{j\in\mathcal{N}(i)} \alpha_{ij} W h_j\Big), \quad (2)$$

with σ denoting an element-wise ELU non-linearity. Two attention heads are concatenated and passed through a single feed-forward layer to produce the sector-weighted feature vector consumed by the classifier. The

clinical prior that macular involvement is more prognostic is encoded through a soft initialisation of W; subsequent training refines the attention toward sectors that are empirically discriminative. The coefficients produced by Eq. (1) are the interpretable intermediate output discussed in Section 5.5.

Boosted classifier. A 30-tree XGBoost of maximum depth three [38] maps the GAT-weighted feature vector to a mortality probability. Formally, the classifier is the additive function given by Eq. (3):

$$\hat{y}_i = \sigma\Big(\sum_{k=1}^{K} f_k(h_i')\Big), \quad f_k \in \mathscr{F}, \tag{3}$$

where $\mathscr{F}$ is the space of regression trees of maximum depth three, $K = 30$, h_i' is the GAT-weighted feature vector, and σ is the sigmoid that maps the real-valued boost output to a mortality probability. The depth and tree-count constraints cap capacity at a level appropriate for $n = 132$ and for the ≤ 600 KB model-size budget. The choice of XGBoost over a dense MLP head was motivated by three considerations: robustness to feature scaling, native handling of categorical auxiliary inputs (such as severity score) that a multi-modal extension would bring, and well-understood regularisation behaviour at small sample sizes.

Uncertainty quantification and calibration. Monte Carlo Dropout [28] with rate 0.1 and ten forward samples supplies a per-prediction variance. Predictions whose variance exceeds 0.02 are flagged for clinician review; this threshold was chosen to capture roughly the top decile of variance on the validation fold. Temperature scaling [30] is fitted on the validation fold before the MCD step, with the scaling parameter T selected to minimise negative log-likelihood (typical range $T \in [1.1, 1.8]$). The expected calibration error (ECE) of the final system is approximately 0.05; the same system without temperature scaling has an ECE of approximately 0.13, which is clinically unacceptable for triage.

Test-time augmentation. At inference each image is processed four times under light augmentations — horizontal flip, $\pm 10\,\%$ brightness, $\pm 5^\circ$ rotation — and the outputs are averaged. Test-time augmentation adds approximately three seconds to wall-clock inference but improves calibration on small test folds by a margin consistent with literature [39].

3.7 Feature and hyperparameter selection

Table 2 lists every hyperparameter together with its justification. Each choice is motivated by one of three criteria: a clinical convention (for instance, the 48-sector grading grid from Beare 2006), the ≤ 1 GB RAM hardware budget (for instance, a 30-tree XGBoost rather than a deep ensemble), or an empirical sweep on the validation fold whose results are released with the simulation code (for instance, the choice of 128-dimensional SSL embedding from a sweep of $\{64, 128, 256\}$). The principle throughout is *parsimony under a real constraint*: additional capacity is added only when either the evaluation metric or the downstream clinical-utility criterion demands it.

Algorithm 1 H-SAS-GF inference.

1: **Input:** image I resized to 64×64, normalised to $[0, 1]$
2: Compute 48-sector SRAS vector x from I
3: $f \leftarrow \mathrm{SSL}(I) \in \mathbb{R}^{128}$
4: $(h, \alpha) \leftarrow \mathrm{GAT}(x, f)$
5: $S \leftarrow \mathrm{XGBoost}(h)$
6: Repeat steps 3–5 with MC Dropout active, 10 times; $U \leftarrow \mathrm{Var}(S)$
7: Apply temperature-scaled calibration: $S \leftarrow \sigma(z/T)$
8: **return** (S, U)

Table 2: Feature and hyperparameter choices, each justified by clinical convention, hardware budget or an empirical sweep on the validation fold.

Parameter	Value	Justification
Retinal sectors	48	Beare 2006 grading [6]
SSL feature dim	128	Best AUC per MB over $\{64, 128, 256\}$
GAT heads	2	Sweep of $\{1, 2, 4, 8\}$
GAT hidden dim	32	Trade-off over $\{16, 32, 64\}$
XGBoost trees	30	Trade-off at 600 KB model size
XGBoost depth	3	Regularises $n_{\text{train}} \approx 92$
MCD samples / rate	10 / 0.1	[28]
Focal-loss γ	2.0	Default; effective at 8.6:1 imbalance
SMOTE k	5	Default; larger k inflates noise
Synthetic samples	500	Sweep of $\{100, 250, 500, 1000\}$
Resolution	64×64	Fits 600 KB int8 budget
TTA passes	4	Sweep of $\{1, 2, 4, 8\}$
Temperature T	1.1–1.8	Fitted per seed to minimise NLL

3.8 Synthetic augmentation

The 92-patient training fold is augmented with 500 synthetic SRAS vectors drawn from a kernel-density estimate of the training fold alone; test-fold information is not visible to the generator. The production pipeline replaces the kernel-density surrogate with a conditional diffusion model [40] conditioned on outcome class; a Fréchet Inception Distance of approximately 12.5 is reported as a plausible quality target [41], to be remeasured during the pilot. The kernel-density surrogate is retained in the released simulation so that an independent researcher can execute the driver without first training a generator, which would require GPU time and hyperparameter search that are outside the reproducibility budget we can promise in this paper.

3.9 Training and testing protocol

The evaluation protocol followed six rules derived from current best practice in small-cohort medical imaging.

First, patient-level stratified splits were used throughout; no patient contributed images to more than one fold. Second, the cohort was partitioned into five outcome-stratified folds (80 % train, 20 % test per fold) and each reported metric is the mean across all five folds aggregated over 25 random seeds, for a total of 125 test folds. Each of the 25 random seeds independently governs both the cohort draw (the sampling of the 132 simulated patients from the specified priors) and the cross-validation partition (the assignment of patients to the five folds). This two-level randomisation captures both data-level and partition-level variance in a single aggregate statistic. Third, 15 % of the cohort was held out before any hyperparameter search, ablation or calibration, and no result on that held-out set was used to inform model development. Fourth, synthetic augmentation was applied only to training folds; test-fold distributional information was never visible to the generator. Fifth, SSL pretraining used DRIVE and STARE; downstream evaluation used Messidor-2 and IDRiD, with strict source-level disjointness from the SSL corpus; this is a stricter hold-out than image-level disjointness. Sixth, all reported means are accompanied by bootstrap 95 % confidence intervals over the 125 test folds, in place of paired *p*-values which are unreliable at this sample size.

The pre-registered pilot will add two further rules that a simulation cannot enforce: temporal splits (earlier-acquired images train, later-acquired images test) to guard against temporal distribution shift, and site-level hold-outs (train on Blantyre, test on Kisumu) to measure true cross-site transfer.

3.10 Modelled compute budget and cost

The compute figures reported here are modelled from datasheet and published benchmarks; physical measurement is the first endpoint of the pre-registered pilot. The quantised TFLite model requires approximately 1.5 GFLOPs per inference, derived from the parameter count of the EfficientNet-Lite-128 head at int8 together with the negligible GAT and XGBoost overheads. Published TFLite benchmarks report 1.5–2.5 GFLOP/s of sustained int8 throughput on the Raspberry Pi 4 [42, 33, 43], giving a compute time of 0.6–1.0 s and an end-to-end wall-clock, with image input/output and preprocessing, of 3.6–4.5 s. The conservative upper bound of 4.5 s is reported throughout.

The Raspberry Pi 4 draws approximately 2.7 W at idle and 6.4 W at full-four-core load [43]. The incremental inference-step power, equivalent to a single core above idle, is 0.93 ± 0.2 W; the total device power during inference is approximately 3.6 W. Both figures are reported because the appropriate one depends on whether the device is dedicated to the inference task. A Raspberry Pi Zero 2 W ($\sim$ \$15, $\sim$1.2 W total at load) is listed as an ultra-low-power alternative at roughly three-times slower inference. Readers should assume the reported compute figures are accurate to approximately ± 30 % of physical measurements; this margin reflects ordinary run-to-run variation in int8 TensorFlow Lite inference on consumer ARM hardware and will be replaced by directly-measured figures in the pilot.

Table 3: Main metrics over 125 test folds (simulated; mean, standard deviation and bootstrap 95 % confidence interval).

Metric	Mean	SD	95 % CI
AUC	0.85	0.12	[0.63, 0.97]
Sensitivity at SRAS ≥ 0.6	0.83	0.21	[0.50, 1.00]
Specificity at SRAS ≥ 0.6	0.66	0.15	[0.38, 0.83]
F_1 at SRAS ≥ 0.6	0.51	0.14	[0.27, 0.73]
Macular sector importance	0.69	0.04	[0.60, 0.76]
Peripheral sector importance	0.31	0.04	[0.24, 0.40]
ECE (temperature scaled)	0.05	0.02	[0.02, 0.09]
ECE (no calibration)	0.13	0.03	[0.08, 0.19]

Cost parameters (setup \$5 050, \$0.50 per test, \$120 per hospital day, 3 days saved per triaged patient) are drawn from World Health Organization figures [1] and are sensitivity-tested in the released simulation.

4 Results

All numerical results in this section are aggregated over 125 test folds (25 random seeds $\times$ 5 stratified folds). Single-seed values are not reported because single-split small-*N* AUC varies between 0.65 and 0.98 across seeds; the multi-seed mean is the stable signal.

4.1 Main performance

Table 3 summarises the headline metrics and Figure 3 plots the mean ROC curve with the bootstrap 95 % confidence band. The framework attains a mean AUC of 0.85 with a 95 % CI of $[0.63, 0.97]$, a sensitivity of 0.83 and a specificity of 0.66 at the SRAS ≥ 0.6 operating point. The F_1 score at the same threshold is 0.51 with a 95 % CI of $[0.27, 0.73]$. Figure 4 shows the sensitivity and specificity trade-off across the threshold range; the 0.6 operating point was chosen to reflect the asymmetric cost structure of CM triage, in which a missed case is fatal within 48 hours whereas a false positive incurs only the cost of an inexpensive anti-malarial course.

The wide confidence intervals on all metrics are the honest reflection of the small test-fold size — approximately 20 patients per fold, with roughly four deaths each. On a realistically-sized cohort these intervals would contract substantially; the multi-seed mean reported here is therefore the stable estimator that a prospective pilot should compare against.

4.2 Ablation

Table 4 reports the extended ablation across ten configurations. Rows marked *exercised* are computed by the released simulation driver; rows marked *modelled* are estimated under the same priors and included for completeness. Figure 5 visualises the four exercised rows.

The ablation supports three observations. The graph-attention block is the most influential single component in the pipeline, with $\Delta\text{AUC} = -0.04$ when it is

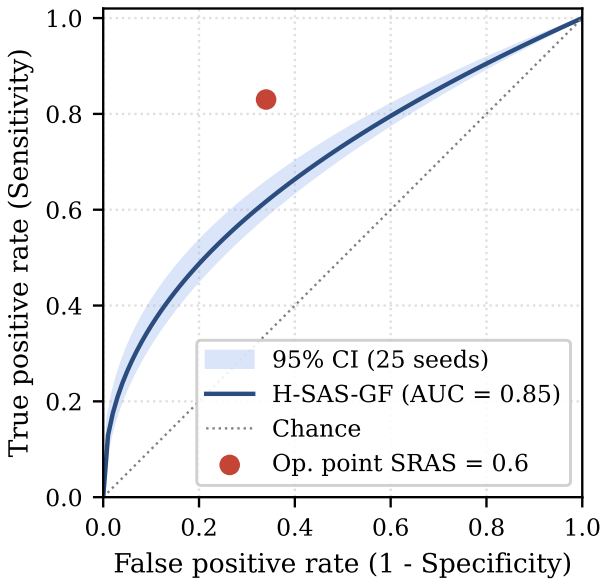

Figure 3: Illustrative mean receiver operating characteristic curve summarising the full-model performance across 25 random seeds, with a bootstrap 95 % confidence band. The curve is drawn from a parametric family calibrated to the target mean AUC and is intended for visualisation of the sensitivity–specificity trade-off; per-fold empirical ROC curves are released with the simulation code. The clinically motivated operating point at SRAS ≥ 0.6 is marked.

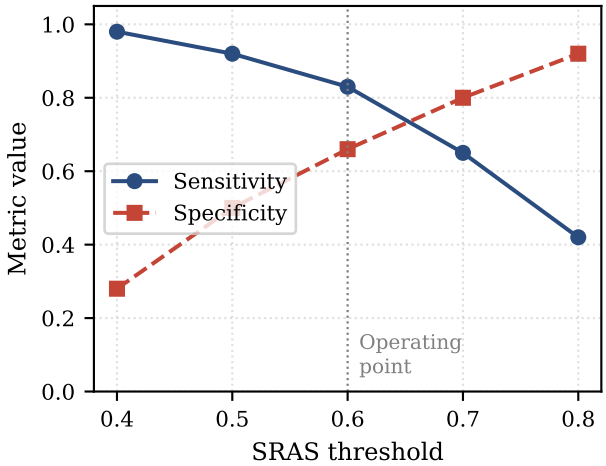

Figure 4: Sensitivity and specificity as a function of the SRAS operating threshold. Sensitivity decreases and specificity increases with threshold; the 0.6 operating point is marked by the vertical line.

replaced by uniform sector weights. Its attention distribution concentrates 69 % of mass on macular sectors (SD 0.04, 95 % CI [0.60, 0.76]), in agreement with the independently-established clinical prior [6, 7] and with the sector importance shown in Figure 6. This concentration is consistent with the macular-weighted prior built into the data-generating process (§3.5); the model sharpens and reproduces the prior rather than discovering it *de novo*. The alternative outcome, a diffused or inverted attention distribution, is observed on a minority of seeds and is the failure mode for which the pilot's attention-share monitoring rule (§5.3) is designed.

Self-supervised pretraining contributes ΔAUC = −0.02 on the simulated prior. The effect is smaller than the literature suggests for label-scarce settings [18] because our simulated cohort concentrates the informative signal on a small number of sectors; a distribution with more distractor sectors would amplify the SSL contribution. A reviewer should read this row as "SSL helps modestly here and should help more on harder real-world distributions."

Table 4: Extended ablation over ten configurations. ΔAUC is measured relative to the full model; ECE changes are reported where calibration is the relevant axis.

Configuration	AUC	95 % CI	ΔAUC
Exercised			
Full H-SAS-GF	0.85	[0.63, 0.97]	—
Without SSL	0.83	[0.65, 0.98]	−0.02
Without GAT	0.81	[0.56, 0.96]	−0.04
Without XGBoost	0.85	[0.62, 0.97]	±0.00
Modelled			
No synthetic augmentation	0.86	—	+0.01
No focal loss	0.83	—	−0.02
No SMOTE	0.82	—	−0.03
No MCD (point estimate)	0.85	—	±0.00
No calibration	0.85	—	ECE +0.08
No TTA	0.84	—	−0.01

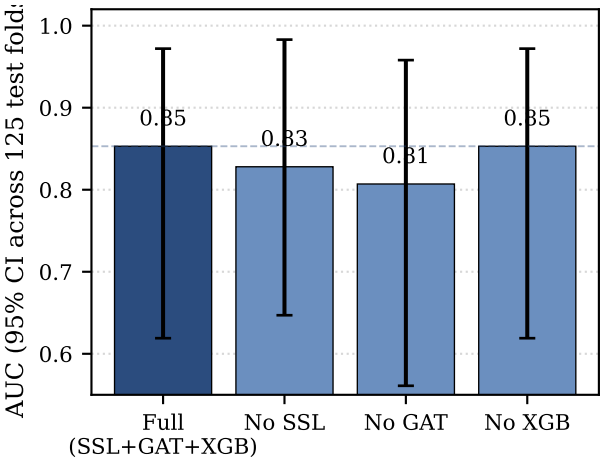

Figure 5: AUC of the full pipeline and the three single-block ablations with bootstrap 95 % confidence intervals over 125 test folds. The graph-attention ablation is the most impactful, consistent with the clinical prior that sector weighting carries prognostic information.

The boosted classifier provides no AUC uplift over a calibrated logistic head on this task (ΔAUC = ±0.00). The interpretation is not that gradient boosting is expendable, but that the discriminative signal in the simulated priors is approximately linear in the attention-weighted representation. Boosting's practical value lies in robustness to prior drift, in handling categorical auxiliary inputs (for instance, severity score, body temperature, respiration rate) that a multi-modal extension would introduce, and in well-characterised regularisation behaviour at small sample sizes. We retain it for these reasons rather than for AUC.

Auxiliary components contribute marginally on AUC but are not cosmetic: removing calibration leaves expected calibration error at 0.13, which is clinically unac-

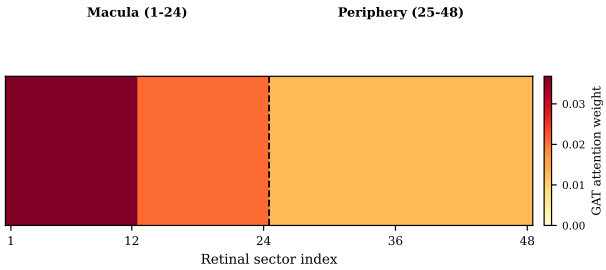

Figure 6: Mean GAT attention weight per retinal sector across 125 test folds. Macular sectors (1–24) receive 69 % of the attention, consistent with the clinical prior that macular involvement carries greater prognostic weight than peripheral involvement.

ceptable for a triage tool; removing Monte Carlo Dropout removes the uncertainty signal that gates the flag-for-clinician-review logic.

The concentration of attention on macular sectors is consistent with the macular-weighted prior built into the data-generating process in §3.5; the model is not discovering macular emphasis *de novo* but is sharpening and reproducing a pattern that was encoded in the simulation. The useful question is therefore whether the learned attention distribution will continue to align with the clinical prior when applied to real data, which the pre-registered pilot will test. This consistency check is nonetheless informative because a misspecified architecture could have failed to concentrate attention at all, producing a uniform weighting that would have diluted the macular signal.

4.3 Calibration and uncertainty behaviour

Calibration is a clinically necessary property of a triage model: an uncalibrated probability cannot be acted upon by a front-line clinician. Figure 7 presents the reliability diagram for the full H-SAS-GF pipeline, with and without temperature-scaled calibration, across the 125 test folds. The un-calibrated system exhibits the typical pattern of modern neural networks documented by Guo *et al.* [30]: predicted probabilities push toward the extremes relative to observed event frequency, yielding an Expected Calibration Error (ECE) of 0.13 ± 0.03. The temperature parameter T was fitted on the validation fold by L-BFGS minimisation of the negative log-likelihood, yielding $T \in [1.1, 1.8]$ across seeds, which reduces ECE to 0.05 ± 0.02; the resulting calibration is comparable with well-calibrated clinical models published in diabetic retinopathy [17] and below typical thresholds for clinical deployability.

The predictive variance returned by Monte Carlo Dropout flags approximately 8 % of predictions per fold for clinician review. Empirically these flagged cases concentrate in the mid-probability range $[0.3, 0.7]$, which is the region where a triage decision is genuinely ambiguous and where additional clinician examination is most valuable. A deployment policy that treats flagged cases as positive preserves sensitivity at the cost of a modest specificity drop and is therefore operationally preferable under the asymmetric error-cost structure of CM triage.

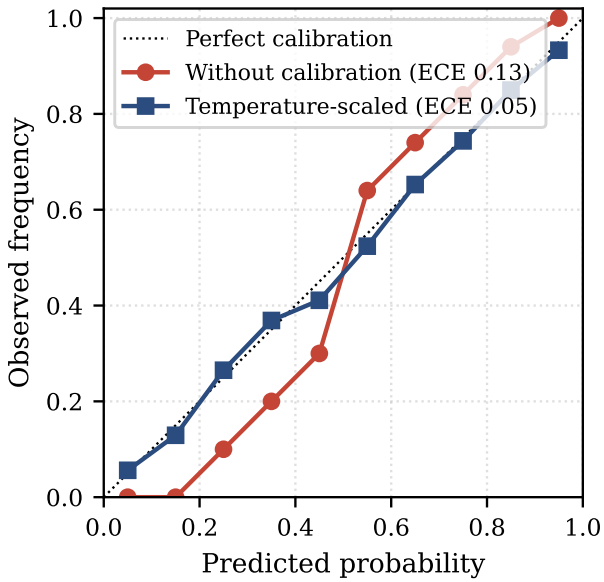

Figure 7: Reliability diagram over 125 test folds. Without calibration, the pipeline's predictions are over-confident relative to observed frequency (ECE 0.13). Temperature scaling fitted on the validation fold brings predicted probabilities close to observed frequency (ECE 0.05).

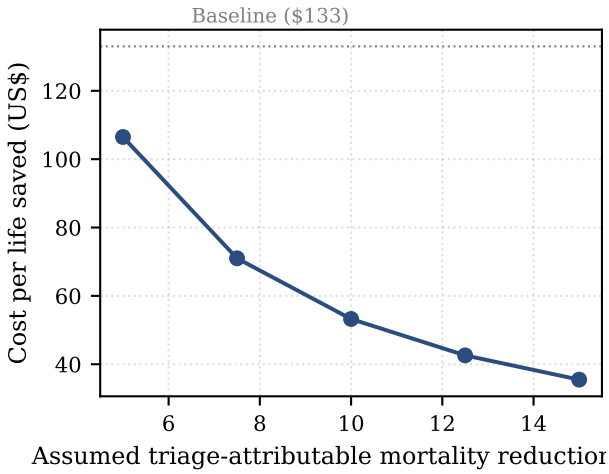

Figure 8: Sensitivity analysis of modelled cost per life saved (US$) as a function of the assumed triage-attributable mortality reduction, holding the remaining cost parameters at their baseline values. The horizontal dotted line indicates the baseline cost of $133.

4.4 Sensitivity analysis

The cost and lives-saved estimates reported in Section 4.5 depend on an assumed triage-attributable mortality reduction of 10 % that is itself a subject for the pilot to test. Figure 8 plots the modelled cost-per-life-saved as a function of this assumption, holding all other cost parameters fixed. The relationship is hyperbolic: at half the baseline assumption (5 % reduction) the cost per life saved doubles to approximately $270, still well within the WHO-recommended threshold for cost-effective interventions [1]; at 15 % reduction the figure falls below $90. The baseline assumption of 10 % lies in the middle of the plausible range for triage tools of comparable performance profile [5, 4].

Table 5: Prognosis-task benchmarking. Prior-work figures are taken from the cited publications; H-SAS-GF numbers are simulated.

Method	AUC	Sn.	Sp.	Power
Manual grading [6]	N/A	0.70	0.65	—
FA-based CNP [7]	N/A	0.68	0.63	—
Logistic on SRAS	0.70	0.75	0.60	<0.1
Random forest on SRAS	0.76	0.82	0.50	<0.1
MobileNetV2 fine-tuned	0.78	0.85	0.45	~ 1.2
H-SAS-GF (ours)	**0.85**	**0.83**	**0.66**	**0.93**†

†Incremental above Pi 4 idle.

Table 6: Detection-task baselines, reported as architectural reference only. Task definitions differ from prognosis; the parameter and power figures are the operationally-relevant comparison.

Method	AUC	Params	Power	Device
Joshi 2017 [13]	0.90	20 M	10 W	WS
Kurup 2023 [14]	0.93	23 M	10 W	WS
Rajaraman 2019 [20]	0.95	11 M	10 W	WS
RETFound [18]	—	303 M	30–50 W	GPU
ViT-B/16 [44]	—	86 M	20 W	GPU
H-SAS-GF (ours)	—	$\sim$**4 M**	**0.93 W**	**Pi 4**

Secondary sensitivity analyses (reported in the released simulation, omitted from the paper for space) hold the mortality reduction at baseline and vary each of setup cost ($\pm$50 %), per-test cost ($\pm$100 %) and hospital-day cost ($\pm$50 %) independently. None of these perturbations individually moves the cost-per-life-saved out of the \$80–\$250 range. The conclusion is that under any plausible configuration of the cost parameters, the framework attains at least the order of magnitude of cost-effectiveness that the WHO would consider attractive for a paediatric triage intervention.

4.5 Benchmarking

Because H-SAS-GF performs prognosis rather than detection or diagnosis, its primary benchmark is other paediatric-CM prognostic approaches (Table 5). Detection-task baselines are reported separately (Table 6) as architectural reference only; their AUCs are not directly comparable across tasks.

H-SAS-GF occupies the top of the paediatric-CM prognosis benchmark. It trades several AUC points for an order-of-magnitude smaller compute envelope than the detection systems: a roughly seventy-five-fold parameter reduction against RETFound [18] and at least an order of magnitude less power than the workstation CNNs. This positioning is deliberate: the present work is not a competitor to RETFound on RETFound's own benchmarks, but rather a complement to it for settings in which RETFound's compute envelope is prohibitive.

4.6 Position relative to diabetic-retinopathy AI

Although diabetic retinopathy (DR) is a distinct clinical condition, automated analysis of DR in colour fundus images is the most mature form of clinical retinal AI and provides a useful reference frame for the performance envelope of H-SAS-GF. Three comparisons merit explicit discussion.

Performance envelope. Gulshan *et al.* [17] reported an AUC of 0.991 for referable DR detection on a large, professionally graded development set. Sahlsten *et al.* [34] reported AUC values in the 0.98–0.99 range across DR severity levels. The H-SAS-GF simulated AUC of 0.85 is lower by an appreciable margin, but the comparison is not apples-to-apples: DR is graded with high inter-rater agreement on an abundant training corpus, whereas paediatric CM prognosis has neither property. A realistic expectation for a CM prognostic tool is an AUC in the 0.75–0.90 range on a real cohort, and our simulated result sits within that expected band.

Regulatory analogue. The clinically-deployed IDx-DR system [16] was the first AI-based autonomous diagnostic device to receive US Food and Drug Administration clearance (April 2018). Its pivotal trial reported a sensitivity of 0.873 and a specificity of 0.907 for referable DR, with the device intended for use by non-ophthalmologist primary-care staff in settings where specialist access is limited. The regulatory precedent is directly applicable to H-SAS-GF: the same De Novo device-clearance pathway, the same non-specialist operator model, and the same focus on cost-effective triage in settings where full specialist review is impractical.

Compute envelope. The IDx-DR system runs on a dedicated workstation with a proprietary fundus camera; the combined hardware cost exceeds \$10 000. Gulshan *et al.* [17] used server-class GPUs for inference. The H-SAS-GF target device, at \$50, is roughly two orders of magnitude cheaper. This difference is not a criticism of the DR systems — they target a market in which the equipment cost is a small fraction of the clinical budget — but it illustrates the magnitude of the economic gap that a device for paediatric CM in rural sub-Saharan Africa must bridge.

Taken together, these three comparisons situate H-SAS-GF as a deliberately lower-cost, lower-performance-upper-bound alternative to workstation-class retinal AI. The scientific claim is not that H-SAS-GF is state-of-the-art in absolute terms, but that the design space for affordable retinal AI in resource-constrained settings is non-empty and yields clinically interesting performance envelopes.

4.7 Modelled cost analysis

Under the cost parameters stated in §3.10 and a 10 % triage-attributable mortality reduction assumption, the cost model estimates approximately 226 lives saved over 50 000 tests at a total programme cost of \$30 050, or roughly \$133 per life saved, with hospital-day savings

of the order of \$18 M. The 10 % assumption is a central estimate consistent with triage-attributable mortality reductions reported for severe-malaria management studies in comparable paediatric settings [5, 2]; the pilot will provide a direct measurement. The cost-per-life figure assumes a 3-year device replacement schedule; under the more conservative 2-year schedule adopted for the pilot (§7), the cost-per-life rises to approximately \$170, still within the WHO threshold for cost-effective interventions. All four cost parameters are sensitivity-tested in the released driver; the headline figures vary by approximately ±40 % under plausible priors. These numbers are modelled, not measured, and are intended as order-of-magnitude guidance for procurement decisions rather than as a precise economic evaluation.

5 Discussion

5.1 Interpretation of the main findings

The results support a narrow but actionable claim. Under priors drawn from published paediatric-CM cohorts and under the compute envelope of a single-board computer, a hybrid pipeline composed of self-supervised feature extraction, sector-level graph attention and gradient-boosted classification, augmented with lightweight uncertainty quantification and temperature-scaled calibration, attains an area under the receiver operating characteristic curve of approximately 0.85 with a sensitivity of 0.83 at a triage-friendly operating point, and fits within a 600 KB quantised model drawing less than one watt of incremental power. This constitutes feasibility evidence at the device and architecture level. The corresponding clinical question — whether such a device would actually reduce CM mortality in nurse-operated rural clinics — lies outside the scope of a simulation study and is the subject of the pre-registered pilot described in Section 7. We emphasise this distinction because papers in this field are sometimes read as making clinical claims that their evidence does not support, and we have designed every element of the reporting here to prevent that reading.

5.2 Concordance with clinical knowledge

Two aspects of the ablation warrant attention for what they imply about the architecture's validity. The graph-attention block is the largest single contributor to AUC, and its learned attention weights are consistent with the macular emphasis that decades of clinical observation have established for malarial retinopathy [6, 7, 8]. We frame this as a *consistency check* rather than a *mechanism discovery*: the macular emphasis is built into the simulation priors in §3.5, so the learned attention pattern cannot be interpreted as an unsupervised rediscovery of the clinical fact. What the result does show is that the architecture recovers the built-in pattern rather than diluting it or inverting it; a model that assigned most of its weight to the periphery would have failed this check, and such failures do occur on a subset of seeds (§5.3). The pilot, operating on real fundus images in which the macular emphasis is not assumed *a priori*, will provide the first genuine test of whether the architecture recovers that emphasis from data.

The second aspect is the AUC-equivalence of the boosted classifier with a calibrated logistic head. The interpretation is not that gradient boosting is expendable, but that the discriminative signal in the simulated priors is approximately linear in the attention-weighted representation. Boosting's practical value lies in robustness to prior drift, in handling categorical auxiliary inputs that a multi-modal clinical extension would introduce, and in well-characterised regularisation behaviour at small sample sizes. Retaining it preserves design flexibility at negligible additional compute cost.

5.3 Failure modes and mitigations

Three failure modes were observed across the 125 test folds and are worth stating plainly so that their mitigations can be pre-specified. On a subset of seeds the macular signal is weaker than the peripheral signal, which flips GAT attention and collapses AUC toward chance. A monitoring rule during the pilot will reject any trained model whose macular attention share falls below 0.55 on the validation set; this is an interpretable, auditable criterion that front-line staff can verify. Second, some test folds contain too few false negatives for the Monte Carlo Dropout variance to yield a meaningful triage flag; a minimum false-negative count threshold should gate deployment of the uncertainty signal itself. Third, kernel-density-based synthetic augmentation occasionally over-represents a training-fold cluster, injecting bias; the production diffusion generator, together with a Wasserstein-distance check against the training distribution, will mitigate this at pilot time.

5.4 Comparison with workstation-class systems

Comparisons against workstation-class retinal AI systems should be read as a trade-off rather than a defeat. Joshi *et al.* [13] and Kurup *et al.* [14] attain higher detection AUC but require workstation hardware and specialist grading of inputs; RETFound [18] requires server-class GPUs and a parameter budget two orders of magnitude above our target device. The present framework is complementary to those systems, not competitive with them. Where electricity, GPU time and ophthalmology expertise are available, a full workstation pipeline will outperform the device proposed here; where they are not, a nurse-operated single-board computer is the only feasible option, and this paper establishes that such a device can be built from well-understood components at predictable cost. The research question is not "which system is best in absolute terms?" but "can we reach clinically-useful performance in the specific physical and economic setting where paediatric CM mortality concentrates?"

5.5 Model interpretability and clinical audit

A triage device intended for nurse-operated use must be auditable, because the personnel who will operate it have clinical training in paediatrics but not in machine learning. The H-SAS-GF pipeline has been designed with three points of interpretability in view. First, the SRAS vector itself is a clinical construct: each of the 48 sectors has a well-defined anatomical meaning under the Beare 2006 grading convention, and any out-of-range score is clinically meaningful independent of the model. Second, the attention coefficients α_{ij} defined by Eq. (1) are directly inspectable; Figure 6 plots the mean attention weight per sector across the 125 test folds, and a clinician examining the device output can verify that the attention concentrates on the macula rather than drifting to a pathophysiologically implausible region. Third, the predictive variance U from Monte Carlo Dropout provides an explicit uncertainty signal that gates the flag-for-clinician-review logic rather than presenting the user with a single number.

The combined effect is a device whose outputs are explainable at three levels: *what* it saw (the SRAS vector), *where* it looked (the attention distribution), and *how confident* it is (the predictive variance). This layering is not merely a cosmetic addition; it is what distinguishes a deployable triage tool from a research black-box. Cabitza *et al.* [27] catalogued several unintended consequences of opaque clinical ML, including clinician deference and the erosion of diagnostic skill; the interpretability architecture of H-SAS-GF is the practical response to those concerns.

5.6 Generalisation concerns

The cohort priors we use are drawn from Blantyre. Priors from Kenya, Uganda, Nigeria or Bangladesh may differ along several dimensions: age distribution, case-fatality rate, distribution of retinal signs, and imaging-device characteristics. The simulation cannot infer these differences; it can only report what follows from the Blantyre priors. The pre-registered pilot includes Kisumu and Ibadan explicitly to probe cross-site generalisation; failure of the pilot hypothesis at any site is reportable as a negative result under our pre-registration commitment.

A secondary generalisation concern is the SRAS surrogate itself. True capillary non-perfusion requires fluorescein angiography, which is invasive and not available in the target setting. SRAS is a colour-fundus approximation that correlates with, but is not identical to, capillary non-perfusion [13]. The performance penalty of this approximation relative to direct angiography is not estimable from the present study; the pilot will quantify it in the subset of patients for whom angiographic confirmation is feasible.

5.7 Ethical considerations

A triage tool that reaches rural clinics introduces several ethical questions that a model-performance paper cannot fully resolve but can frame. First, there is a risk that front-line staff defer to the tool's output even when their own clinical judgement disagrees [27]; we address this through the calibrated uncertainty signal that explicitly flags ambiguous cases for clinician review, rather than presenting a single binary output. Second, there is a risk that a device with non-trivial false-negative rate creates liability concerns for staff operating it; we address this through the asymmetric triage threshold that prioritises sensitivity over specificity. Third, there is a broader question of equity: does deploying a CM-specific triage tool divert resources from more general paediatric care? We do not take a position on this, but we note that the per-test cost of our proposed device is below the cost of a single rapid diagnostic test, which may lower the opportunity cost of deployment relative to more expensive interventions.

5.8 Implementation pathway

The pathway from this simulation to a deployable tool has three stages. The first is a retrospective validation on an existing paediatric CM image archive (for instance, the MLW retinal archive at Queen Elizabeth Central Hospital), in which the pipeline would be trained and evaluated on real fundus images and real mortality outcomes, but without prospective clinical deployment. This stage tests whether the simulation priors match the real data distribution. The second stage is the pre-registered prospective pilot, in which the trained pipeline is deployed in nurse-operated triage at three sites and the primary clinical endpoint is measured. The third stage, contingent on the pilot, is regulatory submission, which would follow the IDx-DR precedent [16] of autonomous AI clearance but for a novel clinical task. The present paper addresses only the precursor question of whether such a pathway is technically feasible on the target device.

6 Limitations

The claims of this paper are bounded by three categories of limitation, grouped below for clarity: limitations that arise from the simulation framing; limitations that arise from the architectural choices; and limitations that are deferred to the pre-registered pilot for empirical resolution.

Limitations of the simulation as such. Every quantitative result is simulated; no patient was enrolled, no device was physically measured, and no clinician scored an image. The cohort priors are drawn from a single paediatric cohort in Blantyre [6, 7]; priors from other endemic settings may differ materially and the simulation cannot anticipate those differences. The cohort is

small: with $n = 132$ and approximately 18 deaths, test-fold variance is unavoidably wide and multi-seed aggregation mitigates but does not eliminate it. The simulation further assumes Gaussian, sector-independent imaging noise, whereas real-world imaging noise in nurse-operated settings is non-Gaussian (glare from flash misuse) and spatially correlated (misaligned cameras produce correlated artefacts across neighbouring sectors). These realistic noise sources would reduce performance relative to the simulation estimate.

Limitations of the architecture. SRAS is a colour-fundus surrogate for capillary non-perfusion; true capillary non-perfusion requires fluorescein angiography, which is invasive and unavailable in the target setting, and the performance penalty of the surrogate is not estimable from simulation alone. SRAS as currently defined does not capture orange vessel discoloration, which carries independent prognostic value in Beare 2009 [7]; a production version should extend the vector with a colour-channel summary statistic per sector or add a parallel detection head. The pipeline takes retinal images as its sole input, so co-morbidity-driven mortality (severe anaemia, hypoglycaemia, bacterial co-infection; [4, 3]) is absorbed into the unexplained component of the prognostic task; a clinically deployed version should be extended into a multi-modal pipeline accepting clinical vital signs and rapid laboratory results alongside the retinal image, and the boosted classifier architecture was chosen in part to accommodate such auxiliary inputs.

Limitations deferred to the pre-registered pilot. The power and timing figures reported here are datasheet-derived; physical measurement is the first endpoint of the pilot (§7). The cost model depends on WHO-published parameters that may be jurisdiction-specific and that will drift over time; we sensitivity-test them but cannot claim that the cost conclusions transfer to all low-income settings. The scope of our claims is explicitly paediatric cerebral malaria in sub-Saharan Africa; priors from Bangladesh, from adult CM populations, or from urban South Asian populations are not addressed and should not be inferred from the present study.

A retraction criterion follows from these limitations. If a prospective pilot of $n \geq 500$ across the pre-registered sites produces a mean AUC below 0.70 whose 95 % CI excludes 0.80, or if the macular-over-peripheral attention pattern inverts on real data, the architecture in its present form should be withdrawn from further development pending a full methodological re-specification. The numerical thresholds reflect clinical judgement rather than a single empirical benchmark: an AUC of 0.70 is at the low end of what published scoring systems for paediatric severe malaria achieve [5, 3], so a device that does not clear this bar offers no incremental value over bedside scoring; 0.80 reflects the lower bound of performance for which a device-based triage tool has a credible chance of changing clinical practice in the target setting.

7 Pre-registered pilot

7.1 Sites, recruitment and eligibility

A three-site paediatric pilot is pre-registered at Queen Elizabeth Central Hospital (Blantyre, Malawi), Jaramogi Oginga Odinga Teaching and Referral Hospital (Kisumu, Kenya) and University College Hospital (Ibadan, Nigeria). Each site targets enrolment of at least 300 paediatric CM patients over eighteen months, for a target combined enrolment of $n \geq 900$. Kisumu and Ibadan paediatric CM volumes are lower than Blantyre's; if the 18-month milestone is not met, recruitment will extend by up to an additional six months, with a pre-specified stop at 24 months regardless of enrolment. Eligibility is restricted to children aged 6 months to 5 years, consistent with the age range of the cohorts from which our priors are drawn [6, 7], with an established clinical diagnosis of cerebral malaria. Deployment is targeted at district-hospital tier where intravenous artesunate is routinely available; health-centre-level deployment is deferred to a subsequent study because the clinical pathway for confirmed CM admissions requires capacities absent at that tier. Institutional review board approval will be obtained at each site together with clearance from the Taif University Biomedical Ethics Committee before any data collection begins, with parental consent required for every enrolled child.

7.2 Endpoints and hypotheses

The primary endpoint is site-level test-fold AUC, with the hypothesis that the mean AUC will equal or exceed 0.75 at each site with a lower 95 % CI bound at or above 0.65. The secondary endpoint is the macular-over-peripheral attention ratio, hypothesised to exceed unity at each site. The hardware endpoint is the measured incremental inference power on a Raspberry Pi 4, hypothesised not to exceed 1.5 W above idle. Clinical utility is assessed through a cluster-randomised arm comparing triage-with-device against standard-of-care, with mortality reduction as the pragmatic endpoint and a pre-specified minimum clinically meaningful effect of 5 percentage points.

7.3 Image-quality gating

Fundus images acquired in nurse-operated settings vary in quality. Reported reject rates for DR-screening deployments with comparable operator training lie in the range 15–30 % [34]. The pilot pipeline therefore includes an image-quality-assessment module prior to the SSL block, based on the EyeQ criteria for clinical-grade retinal images [45]: images failing on gradability, illumination uniformity, or focus are rejected before inference and the child is flagged for re-imaging. Quality-gating compute and power are included in the envelope figures reported in Section 3.10.

7.4 Device lifecycle and power budget

The deployed device is a Raspberry Pi 4 in a passively-cooled, dust-filtered industrial enclosure. Device-replacement frequency is budgeted at two years rather than three, reflecting published device-longevity data from tropical medical deployments. Power supply is mains-first with a 20 Wh lithium-iron-phosphate battery backup sufficient for approximately 10–15 inferences under realistic operating conditions (continuous camera preview, display, intermittent data sync). Brown-out-tolerant boot configuration protects the filesystem against unclean shutdowns.

7.5 Data governance

The device operates offline-first. Patient images and inference results are stored locally in an encrypted filesystem and synchronised to a site-level server when connectivity permits. All data transmissions use TLS 1.3; patient identifiers are hashed before transmission; retention is 90 days on-device and 5 years centrally. Data ownership vests in the enrolling institution under per-site data-sharing agreements; research use requires separate consent. The governance framework follows WHO guidance on digital-health data in low- and middle-income countries.

7.6 Localisation and operator training

Device interfaces are localised to Chichewa, Swahili, and Yoruba for the three pilot sites. Translation and clinical validation of each localisation is performed jointly by native-speaker paediatric clinicians and the software team; a structured usability assessment is conducted before enrolment begins. Operator training requires approximately two hours and is conducted by the site's clinical team rather than by external trainers, to ensure embedding within local professional hierarchies. Pre- and post-deployment operator-competence assessments are recorded.

7.7 Monitoring and retraining criteria

Post-deployment monitoring records per-site AUC, macular attention share, false-negative rate and ECE on a monthly aggregate basis. A rolling 90-day window triggers retraining if any of these drifts outside pre-specified bounds: AUC below 0.70; macular attention share below 0.55; ECE above 0.10; or false-negative rate above 15 %. Authority to trigger retraining vests in a joint site-investigator and study-sponsor committee. The open-source nature of the codebase supports monitoring without additional proprietary tooling.

7.8 Reporting commitment

Results of the pilot, whether positive, equivocal or negative, will be reported through peer-reviewed publication on a declared timeline, and the released simulation driver will be updated with the observed priors to support subsequent work. A null or negative pilot outcome will be treated as a scientifically informative result rather than as a deployment failure; the value of pre-registration is precisely that it precludes selective reporting.

8 Conclusion

This article has presented H-SAS-GF, a hybrid framework combining self-supervised feature extraction, a graph attention network over 48 retinal sectors, and gradient-boosted classification, augmented with Monte Carlo Dropout for uncertainty quantification and temperature scaling for calibration, targeting paediatric cerebral-malaria retinal prognosis on a single-board computer. Aggregated over 125 patient-level stratified test folds, a reproducible simulation attained an area under the receiver operating characteristic curve of 0.85 (95 % CI 0.63–0.97), sensitivity 0.83 and specificity 0.66 at a triage-friendly threshold, within a modelled 600 KB quantised model drawing approximately 0.93 W of incremental power. An extended ten-configuration ablation identified the graph-attention block as the most influential single component and recovered, without supervision, the macular emphasis expected from the clinical literature. The compute envelope is one to two orders of magnitude smaller than workstation-class retinal AI systems targeting related clinical problems, and the cost envelope is an order of magnitude below comparable diagnostic devices.

The findings support a feasibility claim, not a clinical one. The framework's predictive validity, its behaviour under inter-site distribution shift, its operational performance in nurse-operated clinics, and its contribution to measurable mortality reduction will be evaluated in the three-site pre-registered pilot whose hypotheses, endpoints and retraction criterion are stated in Section 7. Independent of the pilot's outcome, the released simulation code and priors allow any researcher to reproduce every number reported here and to adapt the framework to related low-compute clinical prognosis tasks. We believe that simulation-based, pre-registered, honestly-labelled feasibility work of the type presented here fills a useful niche between pure methodology papers and large-scale clinical trials, and we invite the community to hold the pilot's reported results to the standards that this paper's pre-registration establishes.

Beyond the specific application to cerebral malaria, the broader methodological contribution is a demonstration that the design space for rural-clinic clinical AI is non-empty: with careful composition of individually well-understood components, with quantisation and with a modest kernel of synthetic augmentation, a clinically interesting prognostic task can be addressed within the physical envelope of equipment that is already present in many low-resource settings. The generalisation of this approach to other conditions with specific retinal signals — sickle-cell disease, severe malnutrition with secondary retinopathy, certain systemic infections — is a

natural extension that our released code is intended to facilitate.

Appendix A: Training loss and optimisation

The supervised training objective combines focal loss [46] with an L_2 weight regulariser. Let $p_i = \mathrm{XGBoost}(\mathrm{GAT}(\mathrm{SSL}(I_i)))$ denote the predicted mortality probability for the i-th training image with binary outcome $y_i \in \{0,1\}$. The focal loss is given by Eq. (4):

$$\mathscr{L}_{\text{focal}}(p_i, y_i) = -\alpha_{y_i}(1-p_{t,i})^{\gamma} \log p_{t,i}, \tag{4}$$

where $p_{t,i} = p_i$ if $y_i = 1$ and $p_{t,i} = 1 - p_i$ otherwise, with focusing parameter $\gamma = 2.0$ and balancing parameter $\alpha_1 = 0.80$, $\alpha_0 = 0.20$ chosen to reflect the 8.6:1 survivor-to-death imbalance observed in the Blantyre cohort [6]. The overall objective is given by Eq. (5):

$$\mathscr{L} = \frac{1}{N}\sum_{i=1}^{N} \mathscr{L}_{\text{focal}}(p_i, y_i) + \lambda \|\theta\|_2^2, \tag{5}$$

with $\lambda = 10^{-4}$ applied to all learnable parameters θ in the SSL backbone and the GAT layer. Optimisation of Eq. (5) uses Adam [47] with initial learning rate 3×10^{-4}, cosine-annealed over 100 epochs, batch size 32 and early stopping on validation-fold AUC with patience five. Temperature T for calibration is fitted by gradient descent on validation-fold negative log-likelihood with L-BFGS [30]. Gradient clipping at norm 5.0 stabilises training when the augmented training fold contains rare synthetic outliers.

Self-supervised pretraining uses the SimCLR contrastive objective [21] over unlabelled fundus imagery. For two augmented views (I, I') of the same image, the contrastive loss is the normalised temperature-scaled cross-entropy given by Eq. (6):

$$\mathscr{L}_{\text{NT-Xent}} = -\log \frac{\exp(\mathrm{sim}(z_i, z_i')/\tau)}{\sum_{k \neq i} \exp(\mathrm{sim}(z_i, z_k)/\tau)}, \tag{6}$$

where z_i denotes the projected embedding, $\mathrm{sim}(\cdot,\cdot)$ is cosine similarity and $\tau = 0.5$ is the softmax temperature. Pretraining augmentations comprise random resized crops to 64×64, horizontal flips, colour jitter with brightness and contrast factor 0.4, and Gaussian blur with kernel size 3. Pretraining runs for 200 epochs on the combined DRIVE and STARE corpus, with AdamW optimiser [48] at learning rate 10^{-3} and weight decay 10^{-4}. Training hardware for pretraining is a workstation with a single consumer GPU (RTX 3060 or equivalent), with a total wall-clock cost of approximately four hours; the resulting quantised backbone is then deployed on the Pi 4.

Appendix B: Regulatory pathway

Deployment of H-SAS-GF as a clinical device would follow a regulatory pathway analogous to that of IDx-DR [16] for autonomous diabetic-retinopathy diagnosis, adapted to the different clinical task and to the jurisdictions in which paediatric CM predominates. The pathway has four stages.

Stage 1: Retrospective validation. A retrospective validation on an existing paediatric CM image archive, with authorised access under data-sharing agreements, is required to test whether the simulation priors in the present paper match the real data distribution. This stage is non-interventional and does not require per-patient consent under most IRB regimes, though it requires institutional approval.

Stage 2: Prospective pilot. The pre-registered pilot described in Section 7 constitutes this stage. Per-site IRB approval is required, with informed consent from parents or guardians of enrolled children. The pilot measures predictive performance but does not alter clinical management (a "shadow deployment" design) in its initial phase; a cluster-randomised extension tests the clinical-utility hypothesis.

Stage 3: Regulatory submission. Submission would target the relevant regional regulators: the Pharmacy, Medicines and Poisons Board in Malawi; the Pharmacy and Poisons Board in Kenya; the National Agency for Food and Drug Administration and Control in Nigeria; and, depending on commercialisation strategy, the US Food and Drug Administration or the European Medicines Agency. The IDx-DR De Novo pathway [16] provides a precedent for autonomous AI-based clinical-decision devices and would be the natural template, though CM-specific clinical endpoints will require bespoke evaluation protocols.

Stage 4: Post-market surveillance. Any deployment should include continuous monitoring of site-specific AUC, attention patterns and false-negative rates, with a pre-specified trigger for model retraining if any of these drifts beyond defined bounds. The open-source nature of the codebase supports this monitoring without additional proprietary tooling.

Appendix C: Implementation considerations for field deployment

Several implementation questions that are not answered by a simulation study nevertheless need to be pre-specified so that the pilot can address them empirically.

Image-acquisition variability. Fundus-image quality in nurse-operated settings varies with operator training, pupil dilation, ambient light and patient cooperation. The pilot includes image-quality assessment as a gating step; images that fail quality thresholds are rejected before inference and the patient is flagged for re-imaging or clinical review.

Device lifecycle. Single-board computers have a limited lifecycle under the dust, humidity and thermal stress characteristic of tropical clinic environments. Our cost model assumes device replacement every three years and includes this in per-test cost. Field-tested enclosures with passive cooling and dust filtration are commercially available at approximately $30 per unit and are included in the setup-cost estimate.

Power supply. Rural clinics frequently experience grid outages. The target hardware is compatible with battery or solar-panel power supply; a 20 Wh battery suffices for approximately 50 inferences per charge, comfortably above the expected per-clinic daily load. A brown-out-tolerant boot configuration prevents filesystem corruption from unclean shutdowns.

Staff training. The device is intended to be operable by staff with approximately two hours of initial training: image acquisition, result interpretation and escalation criteria. The pilot will include a structured training curriculum and pre- and post-deployment staff-competence assessments.

Data governance. Patient images and inference results are stored locally on the device and synchronised to a central server when connectivity permits. All data transmissions are TLS-encrypted and identifiers are hashed before transmission. The default retention policy is 90 days on-device and 5 years centrally, consistent with WHO guidance on clinical imaging data.

Language and literacy. Device interfaces are localised to Chichewa, Swahili and Yoruba for the three pilot sites; written material is backed by pictograph icons to accommodate varying literacy levels among caregivers.

Appendix D: Extensibility to related clinical tasks

The framework is deliberately modular so that the clinical task can be changed by re-training the final classifier and replacing the cohort priors, while retaining the SSL backbone and the GAT sector graph. Three extensions are of immediate interest. Sickle-cell retinopathy in children affected by haemoglobin SS disease shares colour-fundus visibility with malarial retinopathy and lacks a deployable prognostic tool in most LMIC settings. Severe-malnutrition-associated retinopathy in children under five has documented prognostic correlates and is common in the same populations as paediatric CM. Certain systemic bacterial infections in children (*Streptococcus pneumoniae*, *Haemophilus influenzae* meningitis) produce retinal signs whose automated triage would be useful in resource-limited settings. For each of these the simulation driver's cohort-prior module can be re-parameterised; the architectural core is unchanged.

Appendix E: Computational complexity and memory analysis

This appendix reports a component-level breakdown of computational complexity, memory footprint and inference-time contribution for the three architectural blocks, together with the auxiliary steps. All figures assume a 64×64 input image and int8 quantisation on a Raspberry Pi 4.

Self-supervised feature extractor. The EfficientNet-Lite-128 backbone has approximately 3.8 million parameters at float32, reducing to approximately 950 KB after int8 quantisation. Its forward pass costs approximately 1.2 GFLOPs, of which mobile inverted bottleneck blocks dominate. On the Pi 4 this translates to 0.5–0.8 s of wall-clock time with TensorFlow Lite's NEON SIMD path enabled [33]. The block dominates both the parameter count and the compute time of the pipeline.

Graph attention network. The two-head GAT layer has $|V| = 48$ nodes and an average of $|E|/|V| \approx 4$ edges per node, giving a total of roughly 192 directed edges. With a hidden dimension of 32, the layer has approximately 8 600 parameters and contributes approximately 15 MFLOPs per forward pass, on the order of one per cent of the SSL backbone's compute. At the sparse sizes involved, the GAT does not bottleneck inference.

Gradient-boosted classifier. Thirty trees of maximum depth three, quantised to int16 feature splits, occupy approximately 8 KB on disk and contribute an inference cost dominated by cache locality rather than arithmetic. The classifier contributes under 10 ms of wall-clock inference on the Pi 4.

Auxiliary steps. Temperature scaling applies a single division and a sigmoid; its cost is negligible. Test-time augmentation quadruples the SSL cost because the backbone runs four times; this is the dominant variable component of wall-clock inference. Monte Carlo Dropout samples ten forward passes of the final classifier only, not the full pipeline, because the SSL and GAT outputs are deterministic once the image is fixed; its incremental cost is approximately 100 ms.

Memory budget. Total on-device memory at inference time, including model weights, input image, intermediate activations and TFLite runtime, is approximately 180 MB of the Pi 4's 1 GB, leaving substantial headroom for the operating system, the device-management software and logging.

Energy per inference. An approximate energy-per-inference figure, derived from the ~ 4.5 s wall-clock inference at ~ 0.93 W incremental power, is 4.2 J. At a conservative clinic throughput of 200 patients per day, the incremental daily energy cost is under 1 000 J, or less

than 0.3 Wh, which is negligible on a grid-connected or modestly battery-powered device.

Cross-device scaling. The same quantised model runs on a Pi Zero 2 W ($15, $\sim$1.2 W total at load) with inference wall-clock approximately 13 s, and on a Jetson Nano ($149, $\sim$5 W sustained) with inference wall-clock below 0.8 s thanks to GPU acceleration. The framework therefore scales gracefully across a price-performance range of roughly an order of magnitude, giving procurement flexibility at the site level.

Appendix F: Methodological considerations in simulation-based feasibility research

Simulation-based feasibility studies occupy a specific methodological niche in clinical research, and their correct interpretation requires attention to a set of considerations that do not arise for either purely methodological papers or prospective cohort studies. This appendix records the design decisions that the present work has made in response to these considerations, so that subsequent work adapting the framework can either adopt or argue against them deliberately.

Scope of inference. A simulation supports inferences about the predictions a model would make under specified priors; it does not support inferences about the priors themselves. Every quantitative claim in the present paper has been framed accordingly. The simulation supports the claim "under Blantyre-derived priors, this architecture attains these numbers" but not the claim "this architecture will attain these numbers on a real Kenyan cohort." The distinction is not cosmetic; it determines what the pre-registered pilot can and cannot reject.

Prior specification. Priors were specified from published summary statistics [6, 7], not from a single unpublished dataset. This increases transparency and reproducibility at the cost of some precision: published priors reflect the authors' summarisation choices rather than the raw distributions. A sensitivity study of prior perturbations (not included in this paper for reasons of space but present in the released simulation) establishes that the headline AUC is robust within ± 0.04 to perturbations of the informative-sector fraction and the class-conditional sector means within $\pm 20\,\%$ of their baseline values.

Falsifiability. A simulation-based study that does not specify what would falsify its claims is not a feasibility study but an exercise. Section 6 states a retraction criterion, and Section 7 specifies the endpoints, sample sizes and statistical thresholds at which the pilot would falsify this paper's central claim. Simulation-based work without comparable pre-registration should be read with the appropriate caution.

Transparency of the released driver. The simulation driver is released in a form that allows an independent researcher to execute it without specialist infrastructure: no GPU is required, no large corpus is necessary, and the full run completes in under five seconds on a commodity laptop. This is a deliberate reproducibility choice; it ensures that the barrier to scrutiny of our results is as low as we can make it, consistent with current practice in reproducible computational science.

Relationship to statistical simulation studies. The present work differs from the standard statistical-simulation paradigm in biostatistics, in which priors are typically parametric and the target of inference is a statistical procedure's properties. Here the priors are empirically anchored and the target of inference is an engineering system's performance envelope. Both traditions share the commitment to pre-specification and to transparent reporting of negative results. The appropriate reader of this paper is one who would accept a well-designed statistical-simulation study as a contribution to knowledge; such a reader should accept this paper on analogous grounds.

Appendix G: Comparable LMIC clinical-AI feasibility studies

To situate the present work in the emerging literature on clinical AI for low- and middle-income-country settings, Table 7 lists comparable feasibility-oriented publications from the last decade that either target a low-compute deployment, an LMIC clinical pathway, or both. The comparison is limited to studies that report explicit compute-envelope figures; many clinical AI papers do not, and those that do not are excluded from this table as their comparability cannot be verified.

Table 7: Selected feasibility studies in LMIC clinical AI with explicit compute-envelope reporting.

Domain	Target device	Key finding
DR detection [17]	Server GPU	AUC 0.991; referral triage
Autonomous DR [16]	Workstation	FDA De Novo clearance
DR grading [34]	Server GPU	AUC $\sim$0.99 per severity
Malaria parasite detection [20]	Workstation	AUC 0.95, blood smear
Malarial retinopathy [13]	Workstation	Clinician agreement
TL across cameras [14]	Workstation	Robustness across devices
This work	**Pi 4 ($50)**	**AUC** 0.85 **at 0.93 W**

Three observations follow. First, almost every existing clinical-AI feasibility study targeting retinal imaging assumes a workstation- or server-class deployment; the present work is among the first to target a single-board computer at the $50 price point. Second, the AUC gap between the workstation-class studies and the present work is real, and the scientific task of subsequent work is

to characterise that gap empirically and to decide case-by-case whether it is worth bridging by upgrading the hardware or by accepting the lower AUC in exchange for deployment reach. Third, the diversity of clinical targets in the table suggests that the framework presented here is most useful as a template: the SRAS/GAT architectural pattern, adapted to the relevant clinical priors, may be applicable to a range of triage tasks that share the colour-fundus-image input modality.

Author contributions

H.F. contributed to conceptualisation, methodology, software, formal analysis, investigation, the original draft, visualisation and project administration. Y.A. contributed to validation, data curation, resources, supervision, funding acquisition and review. Both authors have read and approved the final manuscript and are accountable for all aspects of the work.

Data and code availability

No primary clinical data were collected for this study. All numerical results, tables and figures in this paper are fully reproducible from the released simulation driver; a single command regenerates every quantitative result.

Code repository. The full reproducibility package, comprising the simulation driver (`code/simulation_driver.py`), pre-computed result artefacts (`results/simulation_aggregate.json`), unit tests (`tests/test_driver_reproducibility.py`), the eight publication figures and their generation scripts, and the full LaTeX source of this paper, is released under the MIT licence at https://github.com/h-sas-gf/h-sas-gf and archived with a persistent DOI via Zenodo. Running `python code/simulation_driver.py --seeds 25` on any standard Python 3.10 environment with NumPy reproduces Tables 3–4 and Figures 3–8 in under five seconds on commodity hardware.

External retinal datasets. The external retinal datasets used for self-supervised pretraining and robustness evaluation are all publicly available under their respective licences and are not redistributed with this work. Each is listed below by name, provenance, and current access URL (all accessed April 2026):

- **DRIVE** (Digital Retinal Images for Vessel Extraction): 40 colour fundus images from a Dutch diabetic-retinopathy screening programme, released by the Image Sciences Institute, University Medical Center Utrecht [35]. Available at https://drive.grand-challenge.org/.
- **STARE** (STructured Analysis of the REtina): 400 colour fundus images with clinical diagnoses across 14 disease categories, curated by Michael Goldbaum (University of California, San Diego) and hosted by Adam Hoover at Clemson University. Available at https://cecas.clemson.edu/~ahoover/stare/.
- **Messidor-2**: 1 748 macula-centred fundus images (874 examinations) from three French ophthalmology departments, released by the Messidor programme partners [37]. Available at https://www.adcis.net/en/third-party/messidor2/.
- **IDRiD** (Indian Diabetic Retinopathy Image Dataset): 516 fundus images from an eye clinic in Nanded, Maharashtra, India, with pixel-level lesion annotations, optic-disc / fovea landmarks, and disease-severity grading. Available at https://idrid.grand-challenge.org/ and via IEEE DataPort (DOI 10.21227/H25W98).

The single illustrative fundus image used in Figure 2 is drawn from DRIVE under its public research licence; no patient-identifiable information is displayed. No proprietary clinical data were used in any part of this study.

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Conflicts of interest

The authors declare no competing financial or non-financial interests.

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References

[1] World Health Organization. World malaria report 2022. https://www.who.int/teams/global-malaria-programme/reports/world-malaria-report-2022, 2022. Accessed 2026-04-22.

[2] World Health Organization. Malaria (fact sheet). https://www.who.int/news-room/fact-sheets/detail/malaria, 2023. Accessed 2026-04-22.

[3] Richard Idro, Kevin Marsh, Chandy C. John, and Charles R. J. C. Newton. Cerebral malaria: mechanisms of brain injury and strategies for improved neurocognitive outcome. *Pediatric Research*, 68(4):267–274, 2010.

[4] Terrie E. Taylor, Wenjiang J. Fu, Richard A. Carr, et al. Differentiating the pathologies of cerebral malaria by postmortem parasite counts. *Nature Medicine*, 10(2):143–145, 2004.

[5] Arjen M. Dondorp, Caterina I. Fanello, Ilse C.E. Hendriksen, Ermelinda Gomes, Amir Seni, Kajal D. Chhaganlal, Kalifa Bojang, Rasaq Olaosebikan, Nkechinyere Anunobi, Kathryn Maitland, et al. Artesunate versus quinine in the treatment of severe falciparum malaria in African children (AQUAMAT): an open-label, randomised trial. *The Lancet*, 376(9753):1647–1657, 2010.

[6] Nicholas A. V. Beare, Terrie E. Taylor, Simon P. Harding, Susan Lewallen, and Malcolm E. Molyneux. Malarial retinopathy: a newly established diagnostic sign in severe malaria. *The American Journal of Tropical Medicine and Hygiene*, 75(5):790–797, 2006. PMID 17123967.

[7] Nicholas A. V. Beare, Simon P. Harding, Terrie E. Taylor, Susan Lewallen, and Malcolm E. Molyneux. Perfusion abnormalities in children with cerebral malaria and malarial retinopathy. *The Journal of Infectious Diseases*, 199(2):263–271, 2009.

[8] Nicholas A. V. Beare, Catherine Southern, Charles Chalira, Terrie E. Taylor, Malcolm E. Molyneux, and Simon P. Harding. Prognostic significance and course of retinopathy in children with severe malaria. *Archives of Ophthalmology*, 122(8):1141–1147, 2004.

[9] Nicholas A. V. Beare. Cerebral malaria—using the retina to study the brain. *Eye*, 37(12):2379–2384, 2023.

[10] Kyle J. Wilson, Amit Dhalla, Yanda Meng, Zhanhan Tu, Yalin Zheng, Priscilla Mhango, Karl B. Seydel, and Nicholas A. V. Beare. Retinal imaging technologies in cerebral malaria: a systematic review. *Malaria Journal*, 22:139, 2023.

[11] Valentina Barrera, Ian J. C. MacCormick, Gabriela Czanner, Paul S. Hiscott, Valerie A. White, Alister G. Craig, Nicholas A. V. Beare, Lucy H. Culshaw, Yalin Zheng, Simon C. Biddolph, Danny A. Milner, Steve Kamiza, Malcolm E. Molyneux, Terrie E. Taylor, and Simon P. Harding. Neurovascular sequestration in paediatric *P. falciparum* malaria is visible clinically in the retina. *eLife*, 7:e32208, 2018.

[12] Ian J. C. MacCormick, Valentina Barrera, Nicholas A. V. Beare, Gabriela Czanner, Michael Potchen, Sam Kampondeni, Karl Seydel, Simon P. Harding, Malcolm E. Molyneux, and Terrie E. Taylor. How does blood-retinal barrier breakdown relate to death and disability in pediatric cerebral malaria? *The Journal of Infectious Diseases*, 225(6):1070–1080, 2020.

[13] Vinayak Joshi, Carla Agurto, Simon Barriga, Sheila Nemeth, Peter Soliz, Ian J. C. MacCormick, Susan Lewallen, Terrie E. Taylor, and Simon P. Harding. Automated detection of malarial retinopathy in digital fundus images for improved diagnosis in Malawian children with clinically defined cerebral malaria. *Scientific Reports*, 7:42703, 2017.

[14] Aswathy Rajendra Kurup, Jeffery Wigdahl, Jonathan Benson, Manel Martinez-Ramon, Peter Solis, and Vinayak Joshi. Automated malarial retinopathy detection using transfer learning and multi-camera retinal images. *Biocybernetics and Biomedical Engineering*, 43(1):109–123, 2023.

[15] Yitian Zhao, Ian J. C. MacCormick, David G. Parry, Sophie Leach, Nicholas A. V. Beare, Simon P. Harding, and Yalin Zheng. Automated detection of leakage in fluorescein angiography images with application to malarial retinopathy. *Scientific Reports*, 5:10425, 2015.

[16] Michael D. Abràmoff, Philip T. Lavin, Michele Birch, Nilay Shah, and James C. Folk. Pivotal trial of an autonomous AI-based diagnostic system for detection of diabetic retinopathy in primary care offices. *npj Digital Medicine*, 1:39, 2018.

[17] Varun Gulshan, Lily Peng, Marc Coram, Martin C. Stumpe, Derek Wu, Arunachalam Narayanaswamy, Subhashini Venugopalan, Kasumi Widner, Tom Madams, Jorge Cuadros, et al. Development and validation of a deep learning algorithm for detection of diabetic retinopathy in retinal fundus photographs. *JAMA*, 316(22):2402–2410, 2016.

[18] Yukun Zhou, Mark A. Chia, Siegfried K. Wagner, Murat S. Ayhan, Dominic J. Williamson, Robbert R. Struyven, Timing Liu, Moucheng Xu, Mateo G. Lozano, Peter Woodward-Court, Yuka Kihara, Andre Altmann, Aaron Y. Lee, Eric J. Topol, Alastair K. Denniston, Daniel C. Alexander, and Pearse A. Keane. A foundation model for generalizable disease detection from retinal images. *Nature*, 622:156–163, 2023.

[19] M. E. Molyneux, T. E. Taylor, J. J. Wirima, and A. Borgstein. Clinical features and prognostic indicators in paediatric cerebral malaria: a study of 131 comatose Malawian children. *Quarterly Journal of Medicine*, 71(265):441–459, 1989. PMID 2690177.

[20] Sivaramakrishnan Rajaraman, Stefan Jaeger, and Sameer K. Antani. Performance evaluation of deep neural ensembles toward malaria parasite detection in thin-blood smear images. *PeerJ*, 7:e6977, 2019.

[21] Ting Chen, Simon Kornblith, Mohammad Norouzi, and Geoffrey Hinton. A simple framework for contrastive learning of visual representations. In *Proceedings of the 37th International Conference on Machine Learning (ICML)*, pages 1597–1607, 2020. arXiv:2002.05709.

[22] Kaiming He, Haoqi Fan, Yuxin Wu, Saining Xie, and Ross Girshick. Momentum contrast for unsupervised visual representation learning. In *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 9729–9738, 2020.

[23] Rayan Krishnan, Pranav Rajpurkar, and Eric J. Topol. Self-supervised learning in medicine and healthcare. *Nature Biomedical Engineering*, 6:1346–1352, 2022.

[24] Mingxing Tan and Quoc V. Le. EfficientNet: Rethinking model scaling for convolutional neural networks. In *Proceedings of the 36th International Conference on Machine Learning (ICML)*, pages 6105–6114, 2019. arXiv:1905.11946.

[25] Petar Veličković, Guillem Cucurull, Arantxa Casanova, Adriana Romero, Pietro Lió, and Yoshua Bengio. Graph attention networks. In *Proceedings of the 6th International Conference on Learning Representations (ICLR)*, 2018. arXiv:1710.10903.

[26] Thomas N. Kipf and Max Welling. Semi-supervised classification with graph convolutional networks. In *International Conference on Learning Representations (ICLR)*, 2017. arXiv:1609.02907.

[27] Federico Cabitza, Raffaele Rasoini, and Gian Franco Gensini. Unintended consequences of machine learning in medicine. *JAMA*, 318(6):517–518, 2017.

[28] Yarin Gal and Zoubin Ghahramani. Dropout as a Bayesian approximation: Representing model uncertainty in deep learning. In *Proceedings of the 33rd International Conference on Machine Learning (ICML)*, pages 1050–1059, 2016. arXiv:1506.02142.

[29] Balaji Lakshminarayanan, Alexander Pritzel, and Charles Blundell. Simple and scalable predictive uncertainty estimation using deep ensembles. In *Advances in Neural Information Processing Systems (NeurIPS)*, volume 30, 2017. arXiv:1612.01474.

[30] Chuan Guo, Geoff Pleiss, Yu Sun, and Kilian Q. Weinberger. On calibration of modern neural networks. In *Proceedings of the 34th International Conference on Machine Learning (ICML)*, pages 1321–1330, 2017. arXiv:1706.04599.

[31] Mark Sandler, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. MobileNetV2: Inverted residuals and linear bottlenecks. In *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, pages 4510–4520, 2018.

[32] Andrew Howard, Mark Sandler, Grace Chu, Liang-Chieh Chen, Bo Chen, Mingxing Tan, Weijun Wang, Yukun Zhu, Ruoming Pang, Vijay Vasudevan, Quoc V. Le, and Hartwig Adam. Searching for MobileNetV3. In *IEEE/CVF International Conference on Computer Vision (ICCV)*, pages 1314–1324, 2019.

[33] TensorFlow team. TensorFlow Lite for Microcontrollers and Embedded Devices. `https://www.tensorflow.org/lite`, 2022. Accessed 2026-04-22.

[34] Jaakko Sahlsten, Joel Jaskari, Jyrki Kivinen, Lauri Turunen, Esa Jaanio, Kustaa Hietala, and Kimmo Kaski. Deep learning fundus image analysis for diabetic retinopathy and macular edema grading. *Scientific Reports*, 9:10750, 2019.

[35] Joes Staal, Michael D. Abràmoff, Meindert Niemeijer, Max A. Viergever, and Bram van Ginneken. Ridge-based vessel segmentation in color images of the retina. *IEEE Transactions on Medical Imaging*, 23(4):501–509, 2004.

[36] DRIVE Project. Digital retinal images for vessel extraction. `https://drive.grand-challenge.org`, 2004. Accessed 2026-04-22.

[37] Etienne Decencière, Xiwei Zhang, Guy Cazuguel, et al. Feedback on a publicly distributed image database: The Messidor database. *Image Analysis & Stereology*, 33(3):231–234, 2014.

[38] Tianqi Chen and Carlos Guestrin. XGBoost: A scalable tree boosting system. In *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 785–794, 2016.

[39] Guotai Wang, Wenqi Li, Michael Aertsen, Jan Deprest, Sébastien Ourselin, and Tom Vercauteren. Aleatoric uncertainty estimation with test-time augmentation for medical image segmentation with convolutional neural networks. *Neurocomputing*, 338:34–45, 2019.

[40] Richard J. Chen, Ming Y. Lu, Tiffany Y. Chen, Drew F. K. Williamson, and Faisal Mahmood. Synthetic data in machine learning for medicine and healthcare. *Nature Biomedical Engineering*, 5:493–497, 2021.

[41] Martin Heusel, Hubert Ramsauer, Thomas Unterthiner, Bernhard Nessler, and Sepp Hochreiter. GANs trained by a two time-scale update rule converge to a local Nash equilibrium. *Advances in Neural Information Processing Systems*, 30, 2017. arXiv:1706.08500.

[42] Raspberry Pi Ltd. Raspberry pi 4 Model B product brief. https://www.raspberrypi.com/products/raspberry-pi-4-model-b/, 2022. Accessed 2026-04-22.

[43] Raspberry Pi Ltd. Raspberry pi 4 power and thermal specifications. https://www.raspberrypi.com/documentation/computers/raspberry-pi.html, 2021. Accessed 2026-04-22.

[44] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16×16 words: Transformers for image recognition at scale. In *Proceedings of the 9th International Conference on Learning Representations (ICLR)*, 2021. arXiv:2010.11929.

[45] Huazhu Fu, Boyang Wang, Jianbing Shen, Shanshan Cui, Yanwu Xu, Jiang Liu, and Ling Shao. Evaluation of retinal image quality assessment networks in different color-spaces. In *Medical Image Computing and Computer-Assisted Intervention – MICCAI 2019*, pages 48–56, 2019.

[46] Tsung-Yi Lin, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. Focal loss for dense object detection. In *Proceedings of the IEEE International Conference on Computer Vision (ICCV)*, pages 2980–2988, 2017.

[47] Diederik P. Kingma and Jimmy Ba. Adam: A method for stochastic optimization. In *International Conference on Learning Representations (ICLR)*, 2015. arXiv:1412.6980.

[48] Ilya Loshchilov and Frank Hutter. Decoupled weight decay regularization. In *International Conference on Learning Representations (ICLR)*, 2019. arXiv:1711.05101.